ALGEBRA 2 · UNIT 6: EXPONENTIAL & LOGARITHMIC FUNCTIONS

Earthquake & Sound Lab

Use logarithms and exponential models to compare earthquakes, sounds, and ancient bones.

Honors

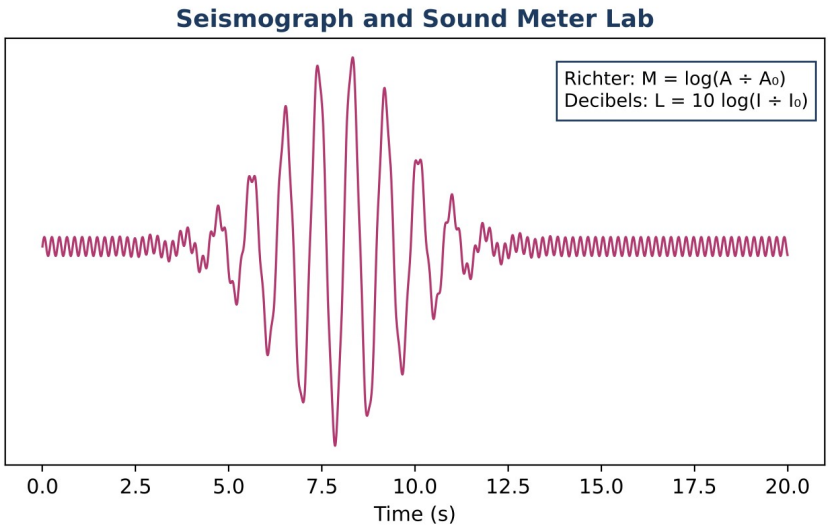
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Scientists measure huge ranges of values: tiny tremors and enormous earthquakes, whispers and jet engines. Logarithmic scales squeeze these ranges into small numbers.

You will compare quakes and sounds, date ancient bones with carbon-14, model cooling, and present what you learn on a safety poster.



Learning Goals

- I can build exponential models from data and convert between forms.
- I can solve equations with different bases.
- I can reason with logarithmic scales.

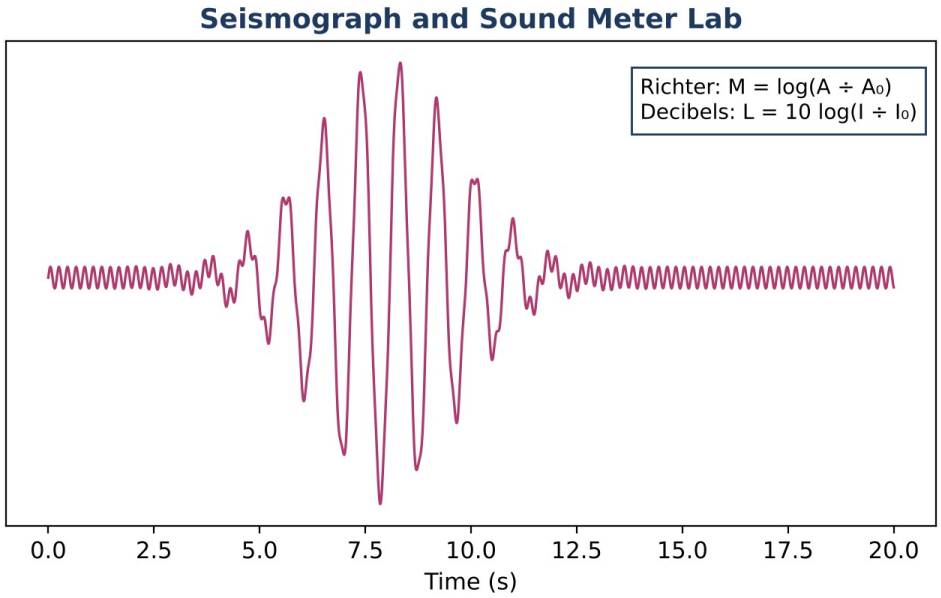
Project Timeline

When	What you will do
Day 1	Scales & Sums
Day 2	Models
Day 3	Poster
Final	Science Safety Poster + reflection

Part 1: Scales & Sums

Day 1 · Use $L = 10 \log(I \div I_0)$.

1. Two speakers each produce 80 dB. Find the combined level. Explain why it is not 160 dB.



2. How many 80-dB speakers would it take to reach 100 dB?

3. Prove $\log(a \div b) = \log a - \log b$ from the exponent rules.

Part 2: Models

Day 2 · Exact and approximate answers.

4. A bacteria culture has 500 cells at t = 2 h and 4,500 at t = 6 h. Find a and b in $N = a \cdot b^t$, then write it as $N = ae^{kt}$.

5. Solve $3^{x+1} = 5^x$.

6. A fossil has 30% of its carbon-14. Find its age, then find the age if the measurement could be off by ±2 percentage points.

Part 3: Poster

Day 3 · Create a poster explaining log scales with real data.

7. Include a model built from two data points.

8. Include an error range for one calculation.

9. Write a public message.

Final Product: Science Safety Poster

Your final product must include:

- Sound-level analysis
- Model from two points in both forms
- Error-range calculation
- Poster

Reflection

R1. Why do equal ratios correspond to equal differences on a log scale?

R2. How does uncertainty in a measurement affect an age estimate?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Comparisons and models are correct and explained in plain language for the public. Goes beyond the minimum with a creative, realistic choice.	Comparisons and models are correct and explained in plain language for the public.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Poster shows the log scales, at least three comparisons with work, and an exponential model solved with logarithms. Polished, organized, and visually engaging.	Poster shows the log scales, at least three comparisons with work, and an exponential model solved with logarithms. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.